Name:

# Final Exam Solutions

**Instructions.** The only tools you are permitted to use are pencils, pens, erasers, a sheet of handwritten notes, and your mind. No electronic devices! You have 2.5 hours. Be sure to sign the Honor Code statement on the last page. Good luck!

**Problem 1** (True-false, 10 points)**.** You will get 1 point for a correct answer and 0 points for blank and incorrect answers. No explanations are required.

(1) The functions $\begin{bmatrix} 2e^{2t} \\ e^t \end{bmatrix}, \begin{bmatrix} e^t \\ 2e^{2t} \end{bmatrix}$ are linearly independent. **T** F

(2) Suppose $x''' = x + x' + \sin(t)$. If $\vec{x} = \begin{bmatrix} x \\ x' \\ x'' \end{bmatrix}$, then $\vec{x}\,' = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \vec{x} + \begin{bmatrix} 0 \\ 0 \\ \sin(t) \end{bmatrix}$. **T** F

(3) Every function $x$ satisfying $x' = t - \cos(x)$ must be increasing for all $t > 1$. **T** F

(4) The initial value problem $y' = 2x + \sin(y)$ and $y(0) = 1$ has a unique solution. **T** F

(5) The initial value problem $(y - 2x - 1)y' = 0$ and $y(0) = 1$ has a unique solution. T **F**

(6) Suppose $A$ is a $2 \times 2$ matrix and $L$ is the linear operator $L\vec{x} = \vec{x}\,' - A\vec{x}$. The null space of $L$ must be 2 dimensional. **T** F

(7) Suppose $A$ is a $2 \times 2$ matrix and $L$ is the linear operator defined by $L\vec{x} = \vec{x}\,' - A\vec{x}$. There must exist an $\vec{x}$ such that $L\vec{x} = \begin{bmatrix} e^{2t} \\ \sin(3t) - \cos(3t) \end{bmatrix}$. **T** F

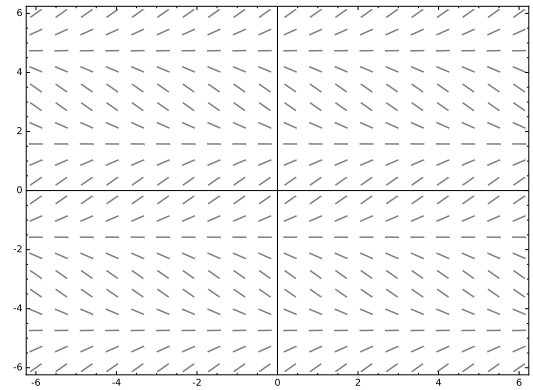
(8) If $A$ is a matrix with characteristic polynomial $-(\lambda - 2)^3$, then $A$ must have 3 linearly independent eigenvectors. T **F**

(9) Suppose $\vec{v}_1, \vec{v}_2$ are linearly independent vectors in $\mathbb{R}^2$. Then there must exist functions $g_1, g_2$ such that $\begin{bmatrix} \sin(2t) \\ \cos(2t) \end{bmatrix} = \vec{v}_1 g_1(t) + \vec{v}_2 g_2(t)$. **T** F

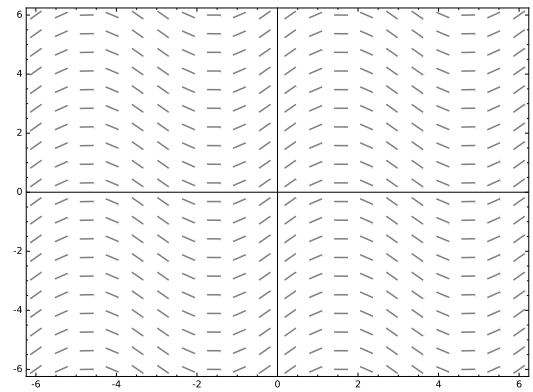
(10) Consider the initial value problem $y' = y$ and $y(0) = 1$, whose solution is $y = e^x$. If we run Euler's method with step size 0.2 to approximate $y(1)$, we end up underestimating the true value of $y(1)$. **T** F

**Problem 2** (3 points)**.** On the left are several first order ODEs. On the right are several slope fields. Match each ODE to its slope field.

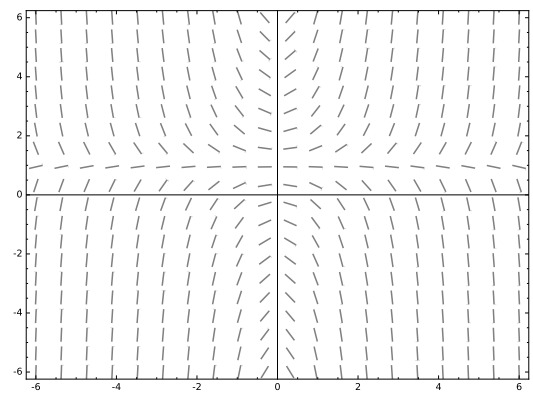
(a) $y' = \cos(x)$



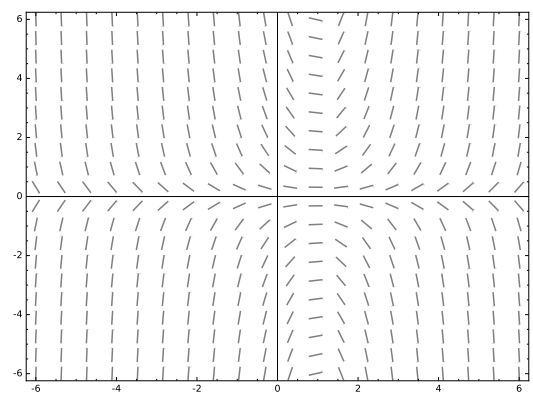
(b) $y' = \cos(y)$



(c) $y' = (x-1)y$



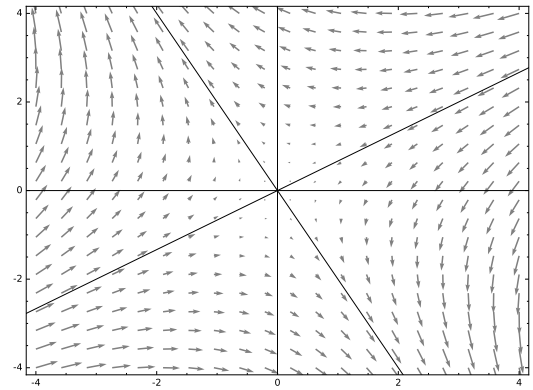
(d) $y' = x(y-1)$



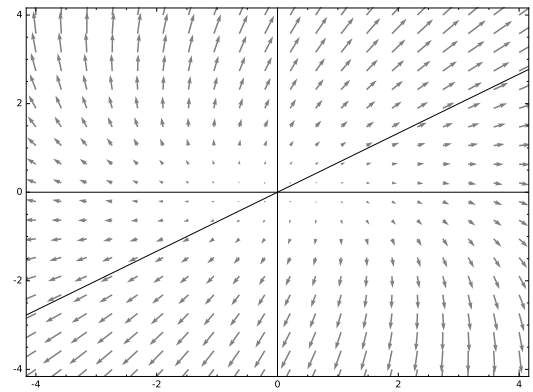
**Solution.** The slope fields belong to the ODEs (b), (a), (d), (c), respectively.

**Problem 3** (3 points)**.** On the left are descriptions of several $2 \times 2$ matrices. On the right are several phase portraits. Match each matrix $A$ with the phase portrait of the system $\vec{x}\,' = A\vec{x}$.

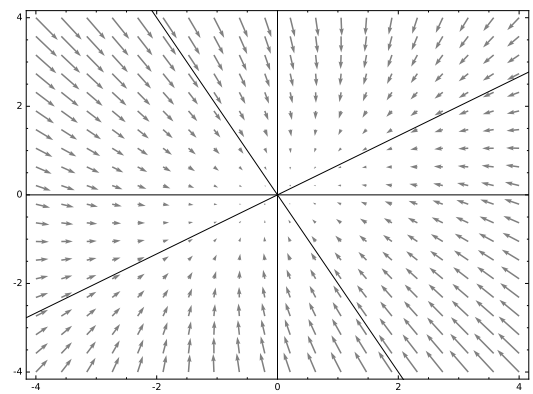
(a) $A$ has complex eigenvalues.



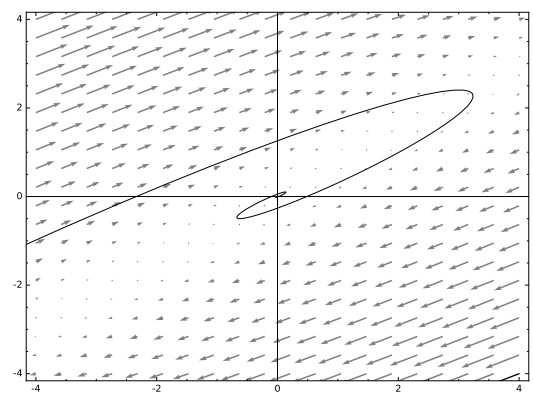
(b) $A$ has negative real eigenvalues.



(c) $A$ has a defective eigenvalue.

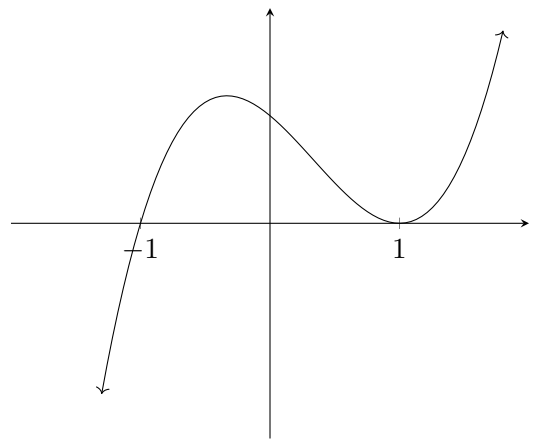


(d) $A$ has two real eigenvalues, one positive and one negative.



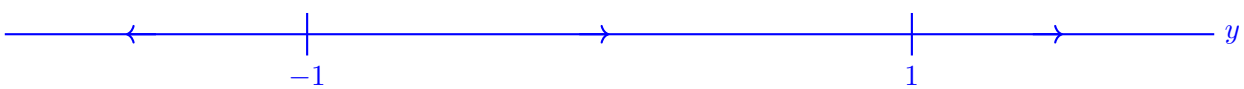
**Solution.** The phase portraits belong to the systems (d), (c), (b), (a), respectively.

**Problem 4** (3 points)**.** Consider the autonomous ODE $y' = f(y)$, where $f$ is the polynomial function whose graph is depicted below.




(a) Draw a phase diagram. Which, if any, of the critical points are stable?

**Solution.** The phase portrait looks like this (except rotated 90° counterclockwise).




There are 2 critical points, at $y = -1$ and $y = 1$. Both are unstable.

(b) If $y$ is a solution to the ODE satisfying $y' = f(y)$ the initial condition $y(0) = -0.5$, what is $\lim_{x\to\infty} y(x)$?

**Solution.** Following the arrows in the phase diagram, we get $\lim_{x\to\infty} y(x) = 1$.

**Note.** Many of you thought that the answer is $+\infty$ (presumably because we just keep following the arrows up and up, across the $y = 1$ critical point), but this is incorrect. If the limit was $+\infty$, that would mean that the function gets unboundedly large; in particular, at some point it would have to actually attain the value $y = 1$. But if it ever did attain the value $y = 1$, then it would have to stabilize at $y = 1$ since $y = 1$ is a critical point, which is a contradiction.

If you've taken real analysis and are unconvinced the above sketch of a proof, I encourage you to try to formalize this argument. If you get stuck, feel free to ask :)

**Problem 5** (2 points)**.** Solve the following initial value problems.

(a) $y' = -3x^2y^2$ and $y(0) = 1$.

**Solution.** Note that $y \equiv 0$ is a solution to the ODE, but it does not satisfy the initial condition. Other solutions to the ODE can be found by separating variables.

$$\begin{aligned} -\frac{dy}{y^2} &= 3x^2\,dx \\ \frac{1}{y} &= x^3 + C \\ y &= \frac{1}{x^3 + C} \end{aligned}$$

Plugging in $x = 0$ and $y = 1$, we find $C = 1$, so the solution is

$$y = \frac{1}{x^3 + 1}.$$

(b) $y' + 3y = 5e^{2x}$ and $y(0) = 2$.

**Solution.** This equation is linear, so we can use the method of integrating factors. An integrating factor is

$$r(x) = e^{\int 3\,dx} = e^{3x}$$

which means that

$$(e^{3x}y)' = e^{3x}y' + 3e^{3x}y = 5e^{3x}e^{2x} = 5e^{5x}.$$

Antidifferentiating with respect to $x$, we get

$$e^{3x}y = e^{5x} + C$$

which means that

$$y = e^{2x} + Ce^{-3x}.$$

Plugging in $x = 0$ and $y = 2$, we get $C = 1$, so the solution is

$$y = e^{2x} + e^{-3x}.$$

**Problem 6** (2 points)**.** A 1 kg mass is attached to a wall by a spring of stiffness 4 kg/s$^2$ and slides around on a surface with friction $b > 0$. If $x(t)$ denotes the displacement of the object from equilibrium position at time $t$, then

$$x'' + bx' + 4x = 0.$$

For what values of $b$ is the system underdamped? Explain briefly.

**Solution.** The roots of the characteristic equation are

$$r = \frac{-b \pm \sqrt{b^2 - 16}}{2}$$

and the damping depends on the expression under the square root (called the discriminant).

- If $b > 4$, the discriminant is positive and we have overdamping.
- If $b = 4$, the discriminant vanishes and we have critical damping.
- If $b < 4$, the discriminant is negative and we have underdamping.

**Problem 7** (2 points)**.** In each of the following, a matrix $A$ is given, together with some information about its eigenvalues. Use the given information to write down the general solution of the system $\vec{x}\,' = A\vec{x}$.

(a) $A = \begin{bmatrix} 1 & 1 \\ -2 & 3 \end{bmatrix}$ has eigenvalues $2 \pm i$, and an eigenvector corresponding to $2+i$ is $\begin{bmatrix} 1 \\ 1+i \end{bmatrix}$.

**Solution.** A complex-valued solution to the system is given by

$$e^{(2+i)t} \begin{bmatrix} 1 \\ 1+i \end{bmatrix} = e^{2t} \begin{bmatrix} \cos(t) + i\sin(t) \\ \cos(t) + i\sin(t) + i\cos(t) - \sin(t) \end{bmatrix}.$$

Taking real and imaginary parts, we find that the general solution is of the form

$$\vec{x} = c_1 e^{2t} \begin{bmatrix} \cos(t) \\ \cos(t) - \sin(t) \end{bmatrix} + c_2 e^{2t} \begin{bmatrix} \sin(t) \\ \sin(t) + \cos(t) \end{bmatrix}.$$

(b) $A = \begin{bmatrix} 5 & -1 \\ 4 & 1 \end{bmatrix}$ has eigenvalue 3 and $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ is an eigenvector. Moreover, $(A - 3I)\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

**Solution.** We have

$$\vec{x} = c_1 e^{3t} \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 e^{3t} \left( \begin{bmatrix} 1 \\ 1 \end{bmatrix} + t \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right).$$

**Problem 8** (3 points)**.** Suppose that you want to solve the following nonhomogeneous differential equations using the method of undeteremined coefficients. Write down the form of the particular solution that you would guess.

(a) $x'' - 3x' - 10x = e^{5t}$.

**Solution.** The characteristic polynomial of the associated nonhomogeneous equation is $r^2 - 3r - 10 = (r-5)(r+2)$, so the complementary solution is $x = c_1 e^{5t} + c_2 e^{-2t}$, so the guess should be of the form $x_p(t) = ate^{5t}$.

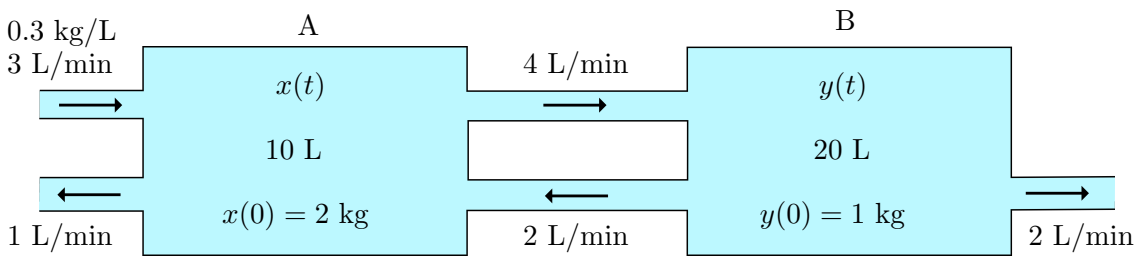
(b) $x'' - 3x' - 10x = t\sin(2t)$.

**Solution.** $x_p(t) = (at+b)\sin(2t) + (ct+d)\cos(2t)$.

(c) $\vec{x}\,' = \begin{bmatrix} 2 & 1 & 0 \\ 0 & -1 & 2 \\ 0 & 0 & 3 \end{bmatrix} \vec{x} + \begin{bmatrix} e^{2t} \\ 5 \\ e^{4t} \end{bmatrix}$

**Solution.** $\vec{x}_p(t) = \vec{a}te^{2t} + \vec{b}e^{2t} + \vec{c} + \vec{d}e^{4t}$

**Problem 9** (3 points)**.** Suppose there are two interconnected tanks A and B with salt water flowing between them as depicted in the diagram below. Suppose further that the input coming into tank A from outside has a constant salt concentration of 0.3 kg/L. Let $x(t)$ and $y(t)$ denote the quantities of salt that are dissolved in tanks A and B at time $t$, respectively. Assume perfect mixing.




(a) Find the matrix $A$ and the vector-valued function $\vec{f}$ such that

$$\frac{d}{dt}\begin{bmatrix} x \\ y \end{bmatrix} = A\begin{bmatrix} x \\ y \end{bmatrix} + \vec{f}.$$

**Solution.** We have

$$\frac{dx}{dt} = -\frac{5x}{10} + \frac{2y}{20} + 0.9 \quad \text{and} \quad \frac{dy}{dt} = \frac{4x}{10} - \frac{4y}{20}.$$

Thus

$$\frac{d}{dt}\begin{bmatrix} x \\ y \end{bmatrix} = \underbrace{\begin{bmatrix} -1/2 & 1/10 \\ 2/5 & -1/5 \end{bmatrix}}_{A}\begin{bmatrix} x \\ y \end{bmatrix} + \underbrace{\begin{bmatrix} 0.9 \\ 0 \end{bmatrix}}_{\vec{f}}$$

(b) What is $\lim_{t\to\infty}\begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$?

**Solution.** One expects that, over time, both tanks will end up with a salt concentration of 0.3 kg/L. This means that tank A will have 3 kg of salt and tank B will have 6 kg of salt, so

$$\lim_{t\to\infty}\begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}.$$

**Problem 10** (Extra credit! 2 points, no partial credit)**.** Ignoring constants of integration, find all functions $f$ such that

$$\left(\int f(x)\,dx\right)\left(\int \frac{1}{f(x)}\,dx\right) = -1.$$

**Solution.** Here's a sketch: take derivatives until you arrive at a nonlinear second order ODE, and then use the substitution $u = f'/f$ to solve that ODE. The answer should be $f(x) = ae^{bx}$ for $a, b \neq 0$.

**Honor code.** If you have neither given nor received any unauthorized aid on this quiz, please write either "HCU" or "Honor Code Upheld" below, and sign your name next to it.

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